Anthropomorphic Robotic Fingers Driven by Cable Pulling and Twisted String Actuation: Design and Experimental Evaluation

Bidipta Rana[1] and Bhivraj Suthar[2]

[1] Shri Ramdeobaba College of Engineering and Management, Nagpur, Maharashtra, India,
bidipta2905@gmail.com,
[2] Indian Institute of Technology Jodhpur, Jodhpur, India
bhivraj@iitj.ac.in

Abstract. This paper presents the design and experimental evaluation of anthropomorphic robotic fingers actuated using two distinct mechanisms: Cable Pulling and Twisted String Actuation (TSA). The two finger prototypes were developed to emulate human-like motion and efficient force transmission, targeting applications in prosthetic and robotic hands. The Cable Pulling mechanism employs tendon-inspired routing for joint articulation, while the TSA-based design leverages motor-driven string twisting to enable compact and energy-efficient actuation. The study details the mechanical design, fabrication process, and control strategies for both systems. A series of comparative experiments were performed to assess range of motion, actuation speed, and energy consumption. The results reveal the strengths and limitations of each approach, providing valuable insights for selecting appropriate actuation methods in bio-inspired robotic fingers.

Keywords: Bio-inspired robotics, Actuations, Cable driven systems

1 Introduction

The development of anthropomorphic robotic fingers has gained significant attention in recent years due to their potential applications in prosthetics, assistive devices, and dexterous robotic hands. Mimicking the complexity and functionality of the human finger remains a challenge, particularly in achieving compact, energy-efficient, and high-force actuation within constrained form factors. A critical aspect of such designs lies in the actuation mechanism, which directly impacts the finger's performance in terms of motion range, responsiveness, and energy consumption. [1], [2], [3]

Traditional actuation strategies such as cable-driven tendon mechanisms have been widely used in robotic fingers due to their simplicity, compliance, and bio-inspiration. These systems replicate the function of biological tendons and muscles, offering natural joint articulation. However, they often require bulky routing

and careful tension control. In contrast, Twisted String Actuation (TSA) has recently emerged as a compact and lightweight alternative, capable of producing high linear contraction and force amplification through the twisting of strings driven by a motor. TSA systems are particularly attractive for applications requiring energy efficiency and mechanical simplicity. [4], [5], [6].

Despite individual advantages, a comparative understanding of these two actuation approaches, Cable Pulling vs. TSA in the context of anthropomorphic finger design is limited. This study addresses this gap by designing, fabricating, and experimentally evaluating two robotic finger prototypes, each utilizing one of the actuation techniques. The goal is to assess their respective strengths, limitations, and suitability for future prosthetic or robotic hand systems.

2 Related Work

Given the need to replicate human hand functions for prosthetics, assistive devices, and humanoids, extensive research has focused on robotic finger design, exploring diverse actuation methods for dexterous, bio-inspired motion. The distinct anatomical components of the human hand allow for precise movements and grasp postures. The primary objective is to replicate hand movements by simulating joints' degrees of freedom (DOFs) and range of motion (ROM) and carrying out various tasks [7]. The kinematic and geometric features of the human hand serve as the key benchmark for designing humanoid prosthetic fingers, which should closely match natural size, shape, and motion to support amputees' physical and psychological recovery [8].

The proximal, middle, and distal phalanges of the human finger are joined

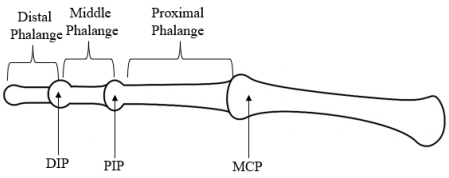


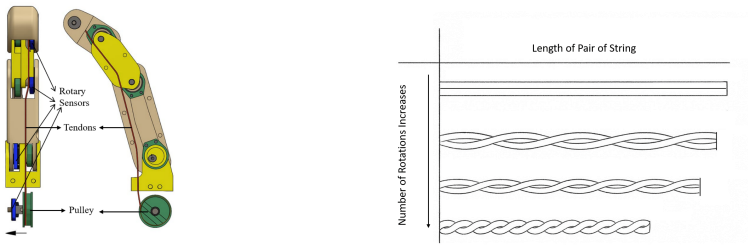
Fig. 1: Basic Architecture of Human Finger depicting phalanges and interphalangeal joints

by the metacarpophalangeal (MCP), Proximal interphalangeal (PIP), and distal interphalangeal (DIP) joints, as shown in Figure 1. The PIP and DIP joints function as single-DOF revolute joints enabling only flexion and extension, while the MCP joint offers two DOFs for flexion/extension and adduction/abduction. Practically, an underactuated anthropomorphic finger with three phalanges and two DOFs balances compliance, cost, and functional adaptability [8].

In a cable-pulling system, a flexible cable runs along the robotic finger's surface, anchored at the fingertip and guided to transmit force through all joints.

Instead of using multiple pulleys, a single motor-driven pulley placed remotely pulls the cable to flex the joints together. Finger extension is achieved either by reversing the motor or using passive elements like springs. This offsite setup reduces on-finger weight, simplifies design, and mimics the human tendon system, balancing mechanical simplicity with practical dexterity [10](Fig: 2a). The fundamental principle of twisted string actuation (TSA) involves attaching one or more parallel strings between a rotating electric motor and the load. As the motor spins, the strings twist and contract, producing linear motion at the output (Fig: 2b). This approach offers a high, though nonlinear, reduction ratio, making it ideal for compact, lightweight applications—here, four TSA units are integrated into one compact design [11].

This research aims to assess and compare the effectiveness of two different actu-



(a) (b)

Fig. 2: (a) Basic Concept of Cable Pulling System: Offsite Actuation setup (rotation motion of pulley is converted to linear motion of string.), (b) Basic Concept of Twisted String Actuation System: As the number of rotations increases the string length decreases.

ation methods—cable-driven actuation and twisted string actuation (TSA)—for operating an anthropomorphic robotic finger. It investigates how each system affects joint motion during flexion and extension while analyzing power use, current draw, and mechanical efficiency. The findings provide valuable insights for selecting and refining actuation methods in robotics and prosthetics.

3 Design and Development of actuation

3.1 Robotic Finger Design

The Cable-Pulling Actuation (CPA) system developed in this study replicates natural finger motion using a tendon-inspired mechanism as shown in Figure 3. A single offsite servo motor (Emax ES08ma II) drives all three joints—MCP, PIP, and DIP—via a cable routed through low-friction channels in the 3D-printed finger segments. As the servo rotates, the cable creates coordinated flexion across the joints, producing a smooth curling motion.

A 2.5 cm pulley on the servo shaft increases cable displacement per rotation, enabling full flexion without requiring high torque. A dorsally mounted elastic

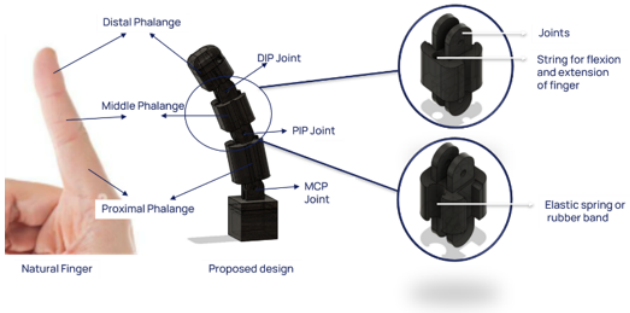


Fig. 3: Proposed CAD design of an anthropomorphic robotic finger: replicating human finger anatomy and integrating mechanical elements for functional movement

strip simulates extensor tendons, passively returning the finger to its extended position when tension is released.

This compact, lightweight system simplifies control, reduces mechanical complexity, and is well-suited for prosthetic applications and robotic hand research.

3.2 Cable-Pulling System

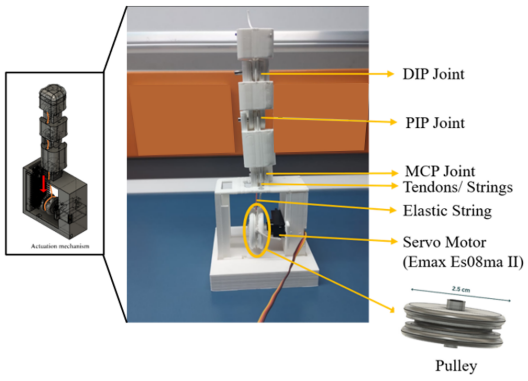


Fig. 4: Experimental setup of the CPA system for evaluating various metrics. The 3D-printed finger model with a fixed servo motor, pulley, and string anchored at the fingertip.

The finger design uses an offsite actuation system, where motion is transmitted via cables from a remote servo motor rather than an embedded actuator as shown in Figure 4. Cables run through low-friction channels in each 3D-printed segment, converting servo rotation into linear tension for coordinated flexion of

the MCP, PIP, and DIP joints. A 2.5 cm pulley on the servo enhances cable displacement, enabling full curling motion with a single actuator.

A dorsally placed elastic strip simulates extensor tendons, passively returning the finger to its extended position when the cable is released. This setup achieves natural, synchronized motion while keeping the system lightweight, compact, and energy-efficient, making it well-suited for prosthetics and robotic hand applications.

3.3 Twisted String Actuation System

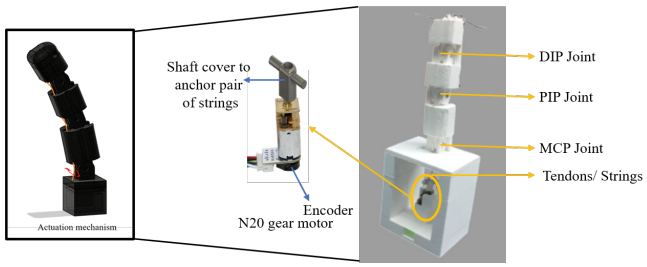


Fig. 5: TSA system experimental setup showing the gear motor with encoder used for performance evaluation.

The Twisted String Actuation (TSA) system offers a compact and efficient method for transmitting force, ideal for lightweight and space-constrained robotic applications. TSA converts rotational motion from a motor into linear displacement by twisting flexible strings, effectively mimicking tendon-driven finger flexion.

In this actuation, two strings symmetrically attached to a rotating shaft twist around each other, shortening in length and pulling the robotic finger, as shown in Figure 5. An N20 gear motor with an integrated encoder drives the system, offering high torque and precise motion control. A custom housing stabilizes the motor, reducing noise and misalignment.

To enable finger extension, a passive return mechanism using an elastic string stores and releases energy, eliminating the need for an additional actuator. This approach results in a lightweight, energy-efficient, and biomimetic finger actuation method—well-suited for robotic hands requiring compact and responsive control.

3.4 Experimental Setup

To compare the performance of cable-pulling and twisted string actuation (TSA) systems, both mechanisms were implemented on identical anthropomorphic robotic finger prototypes, ensuring consistent joint geometry and material properties. The cable-driven finger used an Emax ES08ma II servo motor with a 2.5 cm pulley and a high-tensile cable routed through integrated channels. In contrast, the TSA version employed an N20 gear motor with an encoder, mounted at the base, twisting paired strings for flexion and relying on a passive elastic element for extension.

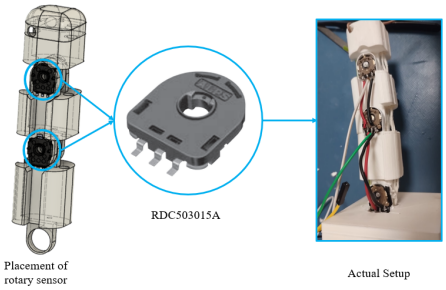


Fig. 6: Placement and Routing of RDC503015A rotary position sensor (placed at each interphalangeal joints of each finger to measure bending angle of the joints during flexion and extension)

Both systems were tested in no-load conditions, simulating free finger motion, and controlled via a Proportional-Derivative (PD) algorithm. Joint angles were recorded in real time using RDC503015A rotary position sensors, as shown in Figure 6. Performance metrics included range of motion (angular displacement at MCP, PIP, and DIP joints) and power consumption, measured through voltage and current readings. Multiple trials were conducted to ensure statistical reliability, and average values were used for direct comparison between the two actuation methods.

4 Results

The Cable Pulling Actuation (CPA) and Twisted String Actuation (TSA) systems were analyzed using defined metrics to determine their effectiveness for anthropomorphic robotic fingers. Parameters such as range of motion, joint movement sequence, energy efficiency, and mechanical response were assessed through controlled trials. The results reveal distinct strengths and limitations for each approach, providing valuable insights for bio-inspired robotics and prosthetics.

4.1 Joint Angle Flexion and Extension

In analysing the flexion and extension behaviour of the anthropomorphic robotic finger, distinct differences were observed between the cable-pulling and twisted string actuation.

In the cable-pulling system, when the servo motor starts retracting the cable, tension first builds at the base, activating the PIP joint, which begins to flex and reaches a maximum rotation of about 58°. As the cable tightens further, the tension travels along its path to engage the MCP joint, which then flexes up to around 71°. Finally, the force extends to the distal section, prompting the DIP joint to bend last, achieving roughly 34° of rotation. This sequential, base-to-tip activation is driven by the cable's distributed tension, resulting in a smooth, natural curling motion that closely mimics a human finger [Fig: 7a]. During extension, this order reverses: as cable tension is released, the DIP joint extends first, followed by the MCP and finally the PIP, with the dorsally placed

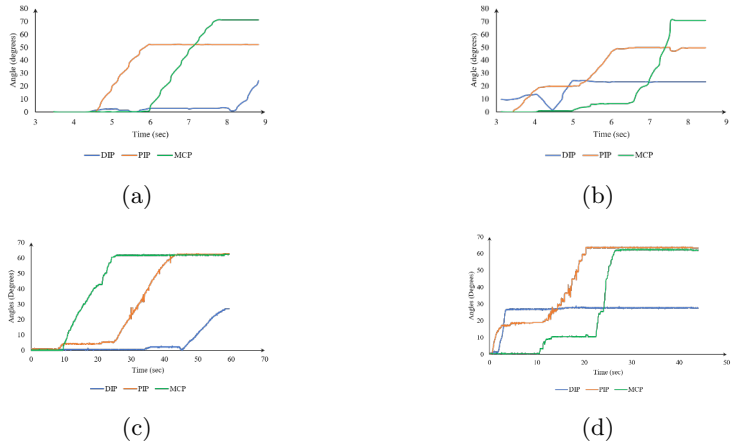


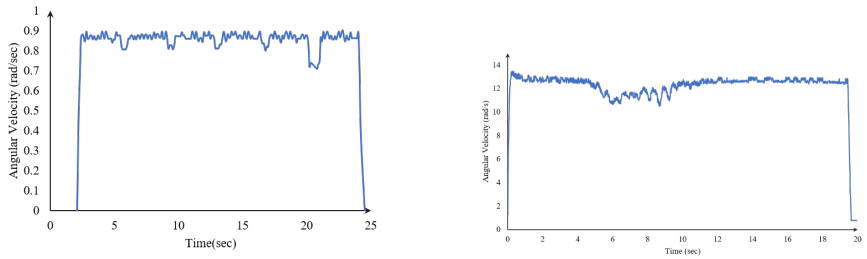
Fig. 7: Comparison of the Cable Pulling Actuation (CPA) and Twisted String Actuation (TSA) systems during flexion and extension based on measured joint bending angles of the DIP, PIP and MCP joints. (a) CPA system during flexion, showing the cable mechanism pulling the fingertip to achieve bending motion. (b) CPA system during extension, illustrating the release of cable tension to straighten the finger. (c) TSA system during flexion, where the twisted string mechanism shortens to bend the finger joints. (d) TSA system during extension, demonstrating untwisting of the string to return the finger to its extended position.

elastic strip assisting the full return to neutral posture [Fig: 7b].
In the twisted string actuation system, finger flexion starts at the MCP joint, which reacts first to the initial twisting as the motor rotates and the strings shorten. The MCP joint curls quickly, reaching about 62°, thanks to its position and alignment with the string path. As the strings twist further, the increasing torsional load activates the PIP joint, which overcomes its resistance and flexes to a peak of 63°, the largest angle due to its central position and favorable lever arm. Finally, the DIP joint responds last, gradually curling as twisting intensifies, achieving a maximum flexion of around 41°[Fig: 7c]. During extension, the motor does not actively drive movement; instead, an elastic return mechanism passively restores the finger. As the motor reverses and the strings untwist, the released torsional tension, combined with the elastic element on the finger's dorsal side, pulls the joints back. The DIP joint extends first, followed by the PIP and MCP, ensuring complete finger straightening [Fig: 7d].

4.2 Velocity vs Time

The Velocity vs Time analysis as shown in Figure 8a, highlights key differences between cable-pulling and twisted string actuation (TSA) systems. The cable-pulling setup shows a smooth, controlled velocity curve with a peak of 0.90 rad/s, reflecting stable acceleration and deceleration—ideal for precise and predictable motion in assistive or research applications.

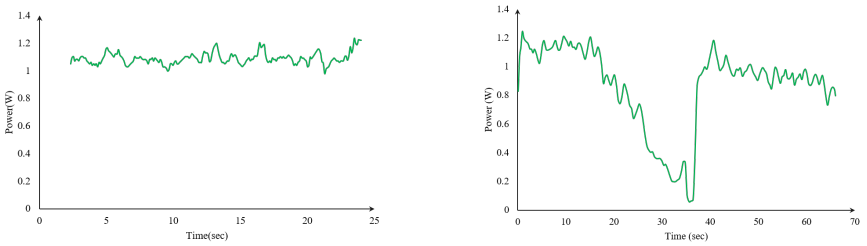
As shown in Figure **??**8b8b TSA system exhibits a sharp, nonlinear velocity spike during flexion, reaching 10.8 rad/s due to the delayed but rapid string contraction. While this enables fast and powerful actuation, it offers less control. During extension, TSA relies on passive elastic return and motor untwisting, resulting in a slower (0.48 rad/s), more irregular velocity profile.



(a) (b)

Fig. 8: Comparison of motor velocities in the Cable Pulling Actuation (CPA) and Twisted String Actuation (TSA) systems during finger flexion and extension movements. (a) CPA system motor velocity profile, showing the velocity characteristics required to pull the string. (b) TSA system motor velocity profile, illustrating the velocity behavior of the robotic finger while twisting and untwisting the string.

4.3 Power Vs Time



(a) (b)

Fig. 9: Comparison of power consumption by the Cable Pulling Actuation (CPA) and Twisted String Actuation (TSA) systems during operation. (a) Power consumption profile of the CPA system, showing the electrical power required to actuate the finger using the cable-driven mechanism. (b) Power consumption profile of the TSA system, illustrating the power used by the twisted string mechanism to perform finger movements.

4.4 Key Findings

The power consumption of Twisted String Actuation (TSA) and Cable Pulling Actuation (CPA) systems was compared over identical actuation cycles. The

Aspect	Cable-Pulley Actuation (CPA)	Twisted String Actuation (TSA)
Joint Angle (Flexion)	MCP: ∼58°, PIP: ∼46°, DIP: ∼34°	MCP: ∼63°, PIP: ∼52°, DIP: ∼41°
Velocity (Peak)	∼0.901 rad/s	∼10.8 rad/s
Current (Peak)	∼0.21 A	∼0.04 A
Power Consumption (Peak)	∼1.18 W	∼0.82 W
Motion Profile	Smooth, symmetric flexion and extension	Fast flexion, slower and passive extension
Ease of Implementation	Simple routing, readily available components	Requires precise string setup and alignment
Durability	High; cables and servos are robust	Strings wear over time, needs periodic replacement
Cost	Low to moderate	Low for components, moderate for tuning & maintenance
Best Suited For	Robotic Hands, Grippers	Prosthetics, soft robotics, energy-efficient systems

Table 1: Comparison of CPA and TSA systems.

CPA system showed a steady but higher power draw, ranging from 1.0 to 1.23 W, with minimal fluctuation, indicating smooth and consistent energy use, as shown in Figure 9a. In contrast, TSA displayed nonlinear power behavior, starting around 0.82–1.18 W, then dropping to 0.4 W due to string twisting dynamics and varying loads, followed by a sharp return to initial values, as shown in Figure 9b.

While TSA exhibited more power fluctuations, it was overall more energy-efficient, achieving greater joint angles, a faster peak flexion speed (10.8 rad/s), and lower peak power usage—making it suitable for compact, low-power applications like prosthetics and soft robotics. Conversely, CPA offered smoother motion, precise control, and more anatomically accurate joint sequencing, ideal for research platforms and structured robotic tasks.

However, TSA requires careful setup, alignment, and periodic string maintenance, whereas CPA is easier to implement and maintain, using standard components. Each system presents clear trade-offs in efficiency, control, and usability, depending on the target application.

5 Conclusion

This work presented the design, development, and experimental evaluation of two anthropomorphic robotic finger prototypes actuated by Cable Pulling and Twisted String Actuation (TSA) mechanisms. Through detailed mechanical design and comparative testing, we demonstrated the feasibility of both approaches for replicating human finger motion in robotic and prosthetic applications. The Cable Pulling mechanism offers reliable and precise control, while the TSA-based

finger provides a compact and energy-efficient alternative with notable force amplification capabilities.

Experimental results highlighted trade-offs between actuation speed, energy consumption, and mechanical complexity. These findings offer critical insights for selecting actuation strategies based on specific application requirements such as portability, power constraints, or dynamic responsiveness.

Future work will focus on integrating these fingers into multi-fingered robotic hands, exploring hybrid actuation approaches, and enhancing control schemes through sensory feedback and machine learning for adaptive grasping in real-world scenarios.

References

1. B. Suthar, Y. Abdulrahman, and Y. Zweiri, 'Robotic Fingers: Advancements, Challenges, and Future Directions-A Comprehensive Review', IEEE Access, vol. 12, pp. 110113–110130, 2024, doi: 10.1109/ACCESS.2024.3440007.
2. B. Kalita, A. Leonessa, and S. K. Dwivedy, 'A Review on the Development of Pneumatic Artificial Muscle Actuators: Force Model and Application', Oct. 01, 2022, MDPI. doi:10.3390/act11100288.
3. S. Min and S. Yi, 'Development of cable-driven anthropomorphic robot hand', IEEE Robot Autom Lett, vol. 6, no. 2, pp. 1176–1183, Apr. 2021, doi:10.1109/LRA.2021.3056375.
4. Suthar, B., Awad, M.I., Seneviratne, L., Zweiri, Y. and Hussain, I., 2024. Design of robotic finger using twisted string actuator with modular passive return rotational joints to achieve high grasping force: Application to wearable sixth finger. Mechatronics, 99, p.103157.
5. I. Gaponov, D. Popov, and J. H. Ryu, 'Twisted string actuation systems: A study of the mathematical model and a comparison of twisted strings', IEEE/ASME Transactions on Mechatronics, vol. 19, no. 4, pp. 1331–1342, 2014, doi:10.1109/TMECH.2013.2280964.
6. I.-W. Park and V. SunSpiral, 'Impedance controlled twisted string actuators for tensegrity robots', in 2014 14th International Conference on Control, Automation and Systems (ICCAS 2014), 2014, pp. 1331–1338. doi:10.1109/ICCAS.2014.6987763.
7. N. Nikafrooz and A. Leonessa, 'A single-actuated, cable-driven, and self-contained robotic hand designed for adaptive grasps', Robotics, vol. 10, no. 4, Dec. 2021, doi:10.3390/robotics10040109.
8. M. Cheng, L. Jiang, F. Ni, S. Fan, Y. Liu, and H. Liu, 'Design of a highly integrated underactuated finger towards prosthetic hand', in 2017 IEEE International Conference on Advanced Intelligent Mechatronics (AIM), Jul. 2017, pp. 1035–1040. doi:10.1109/AIM.2017.8014155.
9. V. Bundhoo, E. Haslam, B. Birch, and E. J. Park, 'A shape memory alloy-based tendon-driven actuation system for biomimetic artificial fingers, part I: Design and evaluation', in Robotica, Jan. 2009, pp. 131–146. doi:10.1017/S026357470800458X.
10. T. Mańkowski, J. Tomczyński, K. Walas, and D. Belter, 'Put-hand—hybrid industrial and biomimetic gripper for elastic object manipulation', Electronics (Switzerland), vol. 9, no. 7, pp. 1–26, Jul. 2020, doi:10.3390/electronics9071147.
11. T. Wuertz, C. May, B. Holz, C. Natale, G. Palli, and C. Melchiorri, 'The Twisted String Actuation System: Modeling and Control', in Mechatronics, IEEE/ASME Transactions on, Jun. 2010, pp. 1215–1220. doi:10.1109/AIM.2010.5695720.